WEEK 1 & WEEK 2

Databricks Intelligence Platform

Updated Features & Deep Dives

- ✥ **Automatic Liquid Clustering (June 2025)**
- ✥ **Auto Loader Type Widening (Runtime 16.4+)**
- ✥ **Auto Loader cleanSource Option**
- ✥ **Serverless Jobs as Default**
- ✥ **10 Advanced Practice Questions**

Supplement to Week 1–2 Study Guide | 2026 Edition

Updated: August 2026

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A **Automatic Liquid Clustering** CLUSTER BY AUTO

■ **NEW (June 2025)**: Automatic liquid clustering intelligently selects clustering keys based on actual query patterns — without requiring manual column specification. This eliminates the guesswork in choosing which columns to cluster.

A1. Manual vs Automatic Liquid Clustering

Feature	Manual CLUSTER BY	Automatic CLUSTER BY AUTO
When available	Databricks Runtime 13.3+	June 2025+ (GA)
Column selection	You specify columns	Databricks AI intelligently selects
Setup effort	Requires analysis	Zero config
Adaptability	Static—must alter for new patterns	Adaptive—evolves with queries
When to use	High-cardinality known columns	Unknown or changing access patterns
Example	CLUSTER BY (customer_id, date)	CLUSTER BY AUTO

■ Automatic Liquid Clustering Syntax

```
-- Manual: explicitly choose columns
CREATE TABLE orders CLUSTER BY (customer_id, order_date) AS
SELECT * FROM raw_orders;

-- Automatic: let Databricks choose (NEW)
CREATE TABLE orders CLUSTER BY AUTO AS
SELECT * FROM raw_orders;

-- ALTER table to switch to auto
ALTER TABLE orders CLUSTER BY AUTO;
```

■ **Key Exam Point:** Automatic clustering is recommended for all UC-managed tables. It requires zero configuration and adapts as query patterns change.

B Auto Loader: Type Widening & cleanSource

Schema Evolution 2.0

NEW Features (Runtime 16.4+, 2026):

• **Type Widening**: Automatically widen types (int→long, float→double) without data rewrite

• **cleanSource**: Auto-delete processed files from source after successful ingestion

B1. Type Widening in Auto Loader Schema Evolution

Type widening allows safe, automatic type promotions when source data changes. For example, if new files contain larger integers (int64) vs. existing int32, Auto Loader automatically widens the schema without rewriting data.

■ Auto Loader Type Widening

```
# Enable type widening with Auto Loader
spark.readStream.format("cloudFiles")
  .option("cloudFiles.format", "parquet")
  .option("cloudFiles.schemaLocation", "/schema/path")
  # NEW: Type widening mode (Runtime 16.4+)
  .option("cloudFiles.schemaEvolutionMode", "addNewColumnsWithTypeWidening")
  .load("/data/path")
  .writeStream
  .option("checkpointLocation", "/checkpoint/path")
  .toTable("target_table")

# Supported type widening:
# int → long
# float → double
# decimal(10,2) → decimal(20,4)
```

B2. Auto Loader cleanSource Option

The cloudFiles.cleanSource option (new in Runtime 16.4) automatically moves or deletes files after they are successfully processed. This simplifies cleanup and avoids re-ingesting data.

■ cleanSource: Automatic File Cleanup

```
# NEW: Auto Loader automatic file cleanup
spark.readStream.format("cloudFiles")
  .option("cloudFiles.format", "json")
  .option("cloudFiles.schemaLocation", "/schema/path")
  # Delete processed files after successful write
  .option("cloudFiles.cleanSource", "delete")
  # OR: move to archive path
  # .option("cloudFiles.cleanSource", "move")
  # .option("cloudFiles.cleanSourceArchiveDir", "/archive/path")
  .load("/landing/json")
  .writeStream
```

```
.mode("append")
.toTable("events_silver")
```

Option	Behavior	Use Case
(default)	Files remain in source after processing	Archive manually; keep raw files as backup
cleanSource: "delete"	Delete processed files immediately	Cost-sensitive; high-volume; trusted pipeline
cleanSource: "move"	Move to archive directory	Compliance; audit trail; later deletion

■ **Exam Focus:** cleanSource is a new Runtime 16.4 feature for cost-optimized ingestion pipelines. Understand the tradeoff: faster cleanup vs. recovery complexity.

C	Serverless Jobs as Default	Compute Evolution 2026

UPDATED (2026): Lakeflow Jobs now defaults to Serverless compute in UC-enabled workspaces. No cluster creation permission required — Databricks manages all infrastructure.

C1. Serverless Jobs in Lakeflow

Aspect	Classic Job Cluster	Serverless Job Compute
Default (UC workspace)	—	■ YES (as of 2026)
Startup time	2–5 min (cold)	< 3 seconds
Infrastructure mgmt	Via cluster config	Fully Databricks-managed
Cost model	DBU/hour (incl. idle)	Per-second execution only
Permissions needed	CREATE CLUSTER	None (UC users can submit)
Photon included	Optional	■ Always enabled
Scaling	Manual config	Automatic

C2. Updated Workload Decision Tree

■ Workload-to-Compute 2026

```
# 2026 Decision: Which compute for my workload?

if workload == "scheduled_etl_job":
    return "Serverless Job (default)"
    # Instant startup, no config, zero idle cost

elif workload == "ad_hoc_sql_dashboards":
    return "Serverless SQL Warehouse"
    # < 3s startup, concurrent users, intelligent scaling

elif workload == "interactive_notebook":
    if team_sharing:
        return "All-Purpose Cluster (shared)"
    else:
        return "Serverless Notebook (new)"
        # Single user, instant compute, zero idle

elif workload == "legacy_complex_config":
    return "Classic Job Cluster (from Instance Pool)"
    # Only if serverless insufficient
```

Scenario	2025 Answer	2026 Answer (NEW)
Nightly ETL: 2TB, 45min	Job Cluster from pool	Serverless (instant start, cheaper)

Team SQL dashboards	SQL Warehouse (classic/pro)	Serverless SQL WH (default)
Data scientist exploration	All-Purpose or single-node	Serverless Notebook (beta/preview)
ML training GPU-heavy	All-Purpose + GPU nodes	Serverless (not yet; use classic)

■■ **Critical Update:** Serverless is now the default recommendation for Lakeflow Jobs in UC workspaces. The old "Job Cluster" answer is now "Serverless Job" unless special config is required.

Section D: Advanced Practice Questions (10 Questions)

■■ Time: 20 minutes (2 min per question)

Q1. A data engineer creates a large managed table in Unity Catalog. They want optimal query performance without manually selecting clustering columns. Which feature should they use?

A) OPTIMIZE ZORDER BY (customer_id, date)
B) CLUSTER BY AUTO
C) ALTER TABLE ... CLUSTER BY (manually selected columns)
D) PARTITION BY (region, year)

Q2. An Auto Loader pipeline processes Parquet files where the amount column changes from INT to BIGINT in newer files. The team uses Databricks Runtime 16.4. Which schema evolution mode automatically handles this type widening?

A) addNewColumns
B) addNewColumnsWithTypeWidening
C) mergeSchema
D) typeWidening (explicit)

Q3. After Auto Loader successfully processes 1000 JSON files from s3://landing/, the data engineer wants to automatically delete those files to save storage costs. Which option enables this?

A) .option("cloudFiles.deleteProcessed", "true")
B) .option("cloudFiles.cleanSource", "delete")
C) VACUUM after the stream completes
D) Run a separate job to delete files

Q4. A team submits a Lakeflow Job in a Unity Catalog-enabled workspace (2026). They do not have cluster creation permission. What compute does Databricks use by default?

A) All-Purpose Cluster (shared)
B) Classic SQL Warehouse
C) Serverless Job Compute
D) Job Cluster (fails: no permission)

Q5. Comparing Serverless Job compute vs classic Job Cluster from an Instance Pool, which statement is TRUE?

A) Instance Pool has faster startup (< 3s vs 2–5 min)
B) Serverless charges zero during idle; classic charges DBU/hour
C) Instance Pool is always recommended over Serverless
D) Both require cluster creation permission

Q6. When should a data engineer still use manual CLUSTER BY instead of CLUSTER BY AUTO?

A) Always use AUTO; manual is deprecated
B) When access patterns are well-known and stable (high-cardinality columns)
C) When the table has < 100TB data
D) Never; AUTO is always superior

Q7. An Auto Loader pipeline uses cloudFiles.cleanSource="move" with cloudFiles.cleanSourceArchiveDir="/archive/". What is the compliance benefit?

A) Files are immediately deleted; no audit trail
B) Files are moved to archive; audit trail preserved for compliance
C) Original source is locked; no re-ingestion possible
D) Archive automatically deletes files after 30 days

Q8. A table was clustered manually with CLUSTER BY (customer_id, region). Six months later, queries now focus on (product_id, order_date). Using CLUSTER BY AUTO avoids which cost?

A) Full table rewrite for ALTER CLUSTER BY
B) Manual Z-ORDER runs
C) Continuous manual OPTIMIZE jobs
D) New Instance Pool provisioning

Q9. Type widening in Auto Loader (Runtime 16.4+) is useful for which scenario?

A) Changing column names in source files
B) Automatically promoting int→long or float→double without data rewrite
C) Converting string to numeric types
D) Removing columns from the schema

Q10. A Lakeflow Job runs every hour on Serverless compute. If the job typically runs 5 minutes, what is the primary cost advantage over a classic Job Cluster?

A) Serverless is always cheaper per DBU
B) Serverless charges only during execution (5 min/hr); classic charges idle time (55 min/hr)
C) Serverless includes auto-VACUUM, reducing storage cost
D) Both cost the same; only speed differs

Answer Key & Explanations

#	Answer	Explanation
1	B	CLUSTER BY AUTO (new, GA June 2025) intelligently selects clustering columns. Manual CLUSTER BY requires prior analysis; Z-ORDER is one-time; partitioning is outdated.
2	B	addNewColumnsWithTypeWidening (Runtime 16.4+) handles automatic type promotion (int→bigint). addNewColumns alone fails on type change; mergeSchema is Delta, not Auto Loader.
3	B	cloudFiles.cleanSource="delete" automatically removes processed files. deleteProcessed does not exist; VACUUM only removes time-travel history, not source files; manual deletion is manual.
4	C	Serverless Job Compute is the 2026 default for UC workspaces; no cluster creation permission required. All-Purpose is for interactive; SQL WH is for SQL only.
5	B	Serverless has zero idle cost (pay per-second execution); classic Job Cluster from pool: ~30s startup (not < 3s, that is Serverless SQL WH); both require different permissions.
6	B	Manual CLUSTER BY is still valid for stable, known high-cardinality columns; CLUSTER BY AUTO is automatic and adaptive, not a full replacement for all scenarios.
7	B	Move to archive preserves audit trail (files exist, dated, immutable); delete removes audit trail; immediate deletion offers no compliance history.
8	A	Changing CLUSTER BY columns manually requires full table rewrite. CLUSTER BY AUTO adapts without explicit rewrites.
9	B	Type widening automatically promotes numeric types (int→long, float→double) without data rewrite. Column rename, type conversion, deletion are handled by other schema evolution modes.
10	B	Serverless billed per-execution second; 5 min × 60 executions/month = 300 min/month. Classic Job Cluster: charged DBU/hour even during idle 55 min/hr = significant cost difference.

Scoring Guide: 9–10 = Exam ready | 7–8 = Good foundation; review Q3, Q10 | 5–6 = Return to Sections A–C; focus on new 2026 features | < 5 = Deep review needed

Updated Exam Readiness Checklist

■ Automatic Liquid Clustering (CLUSTER BY AUTO)
■ Type Widening in Auto Loader schema evolution
■ Auto Loader cloudFiles.cleanSource option (new)
■ Serverless Jobs as default (2026 update)
■ Serverless Job vs Instance Pool vs All-Purpose
■ CLUSTER BY AUTO vs manual clustering tradeoffs
■ cleanSource: "delete" vs "move" for compliance
■ Type promotion (int→long) without rewrite

■ **How to Use This Supplement:**

1. Read Sections A–C (~ 30 min)
2. Take the 10-question quiz (20 min, timed)
3. Compare with answer key
4. Re-study any sections scoring < 70%
5. Cross-reference original Week 1–2 guide for full context

LAB EXERCISES & DEEP DIVES

Hands-On Practice with 2026 Features

Automatic Clustering · Type Widening · Serverless Jobs

LAB A

LAB A: Automatic Liquid Clustering Practice

Prerequisites

- Unity Catalog-enabled workspace
- Databricks Runtime 15.4 LTS or higher
- CREATE SCHEMA and CREATE TABLE permissions

Exercise 1: Create Table with CLUSTER BY AUTO

■ Exercise 1: Create Auto-Clustered Table

```
# Step 1: Setup
USE CATALOG main;
USE SCHEMA default;

# Step 2: Create sample sales table
CREATE TABLE IF NOT EXISTS sales_auto (
  order_id BIGINT,
  customer_id BIGINT,
  product_id BIGINT,
  amount DOUBLE,
  order_date DATE,
  region STRING
) USING DELTA CLUSTER BY AUTO
AS SELECT
  ROW_NUMBER() OVER (ORDER BY RAND()) AS order_id,
  (RAND() * 1000)::BIGINT AS customer_id,
  (RAND() * 100)::BIGINT AS product_id,
  RAND() * 500 AS amount,
  DATE_ADD(CURRENT_DATE(), CAST((RAND() * -365) AS INT)) AS order_date,
  CASE WHEN RAND() < 0.25 THEN 'US-East'
       WHEN RAND() < 0.5 THEN 'US-West'
       WHEN RAND() < 0.75 THEN 'EMEA'
       ELSE 'APAC' END AS region
FROM (SELECT 1 FROM RANGE(100000)) t;

# Step 3: Verify clustering is enabled
DESCRIBE DETAIL sales_auto;
-- Look for: clusteringColumns = [order_id, customer_id] (auto-selected)
```

Exercise 2: Compare Manual vs Auto Clustering

■ Exercise 2: Query Comparison

```
# Create manual cluster table (for comparison)
CREATE TABLE sales_manual (
  order_id BIGINT, customer_id BIGINT, product_id BIGINT,
  amount DOUBLE, order_date DATE, region STRING
) USING DELTA CLUSTER BY (customer_id, order_date)
AS SELECT * FROM sales_auto;

# Run queries to observe optimization benefits
```

```
EXPLAIN FORMATTED
SELECT * FROM sales_auto
  WHERE customer_id = 42 AND order_date >= "2025-01-01";

# Compare with manual clustering
EXPLAIN FORMATTED
SELECT * FROM sales_manual
  WHERE customer_id = 42 AND order_date >= "2025-01-01";

# Both should show data skipping; AUTO may use different columns
```

Observation: Run OPTIMIZE on both tables and compare file statistics. AUTO clustering may choose different columns based on your actual query patterns.

LAB B

LAB B: Auto Loader Type Widening & cleanSource

Exercise 3: Type Widening in Schema Evolution

■ **Exercise 3: Type Widening**

```
# Setup: Create volumes for source and checkpoint
CREATE VOLUME IF NOT EXISTS main.default.al_source;
CREATE VOLUME IF NOT EXISTS main.default.al_checkpoint;

# Create initial Parquet file with INT amount column
from pyspark.sql import Row
initial_data = spark.createDataFrame([
  Row(order_id=1, amount=100),  # amount is INT
  Row(order_id=2, amount=250)
])
initial_data.write.parquet(
  "/Volumes/main/default/al_source/initial/")

# Step 2: Auto Loader with Type Widening (Runtime 16.4+)
(spark.readStream.format("cloudFiles")
  .option("cloudFiles.format", "parquet")
  .option("cloudFiles.schemaLocation",
          "/Volumes/main/default/al_checkpoint/schema")
  # NEW: Enable type widening
  .option("cloudFiles.schemaEvolutionMode",
          "addNewColumnsWithTypeWidening")
  .load("/Volumes/main/default/al_source")
  .writeStream
  .option("checkpointLocation",
          "/Volumes/main/default/al_checkpoint/ckpt")
  .mode("append")
  .toTable("orders_silver"))

# Step 3: Add new file with BIGINT amount
new_data = spark.createDataFrame([
  Row(order_id=3, amount=9999999999)  # BIGINT
])
new_data.write.parquet(
  "/Volumes/main/default/al_source/new_batch/")

# Check schema: amount should be BIGINT (widened automatically)
spark.sql("DESCRIBE orders_silver").show()
```

Exercise 4: cleanSource for Automatic Cleanup

■ **Exercise 4: cleanSource Cleanup**

```
# Auto Loader with automatic file cleanup
(spark.readStream.format("cloudFiles")
  .option("cloudFiles.format", "json")
  .option("cloudFiles.schemaLocation",
          "/Volumes/main/default/al_checkpoint/schema_json")
  # NEW: Delete files after successful ingestion
  .option("cloudFiles.cleanSource", "delete")
  .load("/Volumes/main/default/al_source/json_landing")
```

```
    .writeStream
    .option("checkpointLocation",
            "/Volumes/main/default/al_checkpoint/json_ckpt")
    .trigger(availableNow=True)
    .toTable("events_silver"))

# Alternative: Move to archive instead of delete
# .option("cloudFiles.cleanSource", "move")
# .option("cloudFiles.cleanSourceArchiveDir",
#         "/Volumes/main/default/archive")

# Verify files have been deleted or moved
dbutils.fs.ls("/Volumes/main/default/al_source/json_landing")
# Should be empty after successful run
```

Caution: Using cleanSource="delete" is irreversible. Always test with "move" first to archive, then verify data integrity before switching to "delete" in production.

LAB C

LAB C: Creating Serverless Jobs (Lakeflow)

Exercise 5: Serverless Job Submission

■ **Exercise 5: Serverless Job Submit**

```
# Create a simple transformation notebook and submit as Serverless Job

# Notebook: /Users/me@company.com/transform_orders
# ---
import pyspark.sql.functions as F

orders_df = spark.read.table("main.raw.orders")
clean_orders = (orders_df
  .filter(F.col("order_id").isNotNull())
  .withColumn("order_year", F.year(F.col("order_date")))
  .select("order_id", "customer_id", "amount", "order_year"))
clean_orders.write.mode("overwrite").option("overwriteSchema","true")
  .saveAsTable("main.silver.orders")
print("■ Orders loaded to silver")

# Submit via Python SDK (auto-uses Serverless)
from databricks.sdk import WorkspaceClient

w = WorkspaceClient()
job = w.jobs.create(
  name="Load Orders - Serverless",
  tasks=[{
    "notebook_task": {
      "notebook_path": "/Users/me@company.com/transform_orders"
    },
    "task_key": "load_orders"
  }]
)
print(f"Job ID: {job.job_id}")

# Run job (defaults to Serverless in UC workspace)
run = w.jobs.run_now(job_id=job.job_id)
print(f"Run started: {run.run_id}")
```

Comparison: Serverless vs Classic Job Cluster

Attribute	Serverless Job	Classic Job Cluster	All-Purpose Cluster
Startup time	< 3 seconds	2–5 minutes	1–2 minutes (if running)
Cost model	Per-second execution	DBU/hour (incl. idle)	DBU/hour (must stay running)
Permission needed	None (UC user)	CREATE CLUSTER	CREATE CLUSTER or inherited
Auto-termination	Immediate after job	After job completes	Manual or timeout
Default (2026 UC)	■ YES	—	—
Photon enabled	■ Always	Optional	Optional
Scaling	Automatic	Manual (config)	Manual (instance pool) or auto

Key Insight: Serverless is the 2026 default for Lakeflow Jobs in UC workspaces. Use classic only if you need custom Spark config or libraries not pre-installed in Serverless.

Deep Dive: Feature Tradeoffs & Use Cases

When to Use CLUSTER BY AUTO vs Manual

Scenario	Use AUTO?	Reason
Query patterns unknown	■ YES	AUTO adapts to actual queries
High-volume, variable access	■ YES	AUTO self-tunes without manual ALTER
Stable, high-cardinality column (e.g., customer_id)	■ NO	Manual wins with known patterns
Table < 100TB	■ YES	AUTO sufficient; no ingestion-time clustering needed
Frequent schema/workload changes	■ YES	AUTO evolves; manual requires rewrites

Type Widening Use Cases

Source Change	Type Widening Solves	Example
Integer overflow	int → long without rewrite	order_qty: 100 → 5,000,000
Decimal precision	decimal(10,2) → decimal(20,4)	price: $99.99 → $9,999.99
Floating-point loss	float → double	metric: 1.5 → 1.123456789
No rewrite penalty	Schema evolution without data copy	Multiple widening steps allowed

cleanSource Compliance Matrix

Requirement	delete	move	default (none)
Fast storage cleanup	■	■	■
Audit trail preserved	■	■	■
Recovery possible	■	■ (from archive)	■
Compliance (GDPR/HIPAA)	■ (if delete required)	■ (audit + delete later)	Risk
Cost optimization	■ (immediate)	~	■ (storage stays)

Exam Tips: Updated for 2026

Tip 1: CLUSTER BY AUTO is now a default recommendation

If the exam says "optimize a table without manual config," think AUTO, not Z-ORDER.

Tip 2: Type Widening prevents schema evolution failures

A question about int→long changes with Auto Loader? Answer: addNewColumnsWithTypeWidening

Tip 3: cleanSource is new; test your Qs carefully

Distinguish "delete" (irrev.) vs "move" (audit). Compliance scenarios prefer "move".

Tip 4: Serverless Jobs are 2026 default

Any Lakeflow Job question in UC workspace → assume Serverless unless otherwise stated.

Tip 5: Automatic features = Databricks-managed

AUTO and Serverless shift ops burden from you to Databricks. Exam often tests this shift.